Caner Derici

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Technical Skills

Areas of Experience: Compilers & Language Runtimes · Machine Learning · Distributed Systems
Languages: C++ · Go · Python · Racket · Lean · LLVM · MLIR · MIR/TableGen · CUDA/PTX
Workflows & Productivity: Linux · Neovim · tmux · VSCode · OpenCode · Codex · Git · GH Actions · Obsidian
APIs, DB & Misc: REST · gRPC · GraphQL · OpenAPI · DQLite · MongoDB · PostgreSQL · CI/CD

Education

Ph.D., Indiana University, Bloomington, Computer Science, Compilers — 2015 – 2025
Dissertation: Self-Hosting Functional Programming Languages on Meta-Tracing JIT Compilers

M.Sc., Boğaziçi University, Computer Engineering, Machine Learning — 2012 – 2014

B.Sc., Bilgi University, Computer Science, Compilers — 2005 – 2010

Experience

Amazon Web Services (AWS) — Austin, TX
Applied Scientist - L5 IC · Automated Reasoning Group, Strata Team — *2026 – present*
- Developed and maintained Strata, a unified platform for formalizing language syntax and semantics, and implementing automated reasoning applications.

Canonical USA — Remote, US
SWE II (L5 - IC4) · DSL Compiler · Distributed Orchestration at Scale — *2021 – 2024*
- Designed and developed a custom scripting DSL for (CPU & memory) restricted computations. Built compiler on top of a modified Lua interpreter in Go.
- Developed and maintained Juju, an eventually consistent distributed orchestration system used by ~200 companies globally, capable of handling 1000+-node workloads with 99.9% availability on any infrastructure (Kubernetes or otherwise) across various cloud providers (e.g., AWS, GCE). All in Go.
- Improved reliability and fault tolerance by implementing machine provision services on relational DQLite state back-end, migrating from NoSQL MongoDB (e.g., sample PR).
- Owned client libraries for three years–python-libjuju, Terraform Juju Provider; doubled active users and maintained a steady release cadence.
- Took part in roadmap planning, coordinated cross-team work; mentored junior engineers, and created structured bootcamp process to improve hiring and onboarding.

Indiana University — IN, US
PhD Work · Research Assistant · Course Instructor — *2015 – 2021*
- Independently took an ambiguous, uncharted compiler problem to a working product; built the first-ever tracing JIT compiler that is a full-scale runtime for a self-hosting, production-grade language. Conducted a full performance investigation and designed new optimization algorithms (see PhD dissertation).
- Taught data structures & algorithms, compilers, virtual machines, and domain specific languages.

Asseco SEE Group
Software Engineer — *2010-2012*
- International software company developing virtual payment systems for e-commerce platforms. I developed and delivered 3 virtual point-of-sale projects in 2 years. Used Java, Apache Tomcat, Spring, Mercurial, Jira.

Selected Projects

Pycket: A tracing JIT compiler for full-scale bootstrapping functional language

Built the first tracing JIT capable of running a self-hosting language on a meta-tracing JIT backend, and worked on it for over five years during my PhD. Created a new IR and implemented performance analysis tools, run-time optimizations, and formalisms to improve throughput. Added code-gen for the FFI layer, engines, and meta-continuations for user threads. Used Python, C++, and Racket.

Around the World in Compilers

Designed and implemented a growing collection of experimental compilers, one for each letter of the alphabet. Built shared frontend pipelines targeting LLVM IR, MLIR, NVPTX, and LLVM MIR. Explored MLIR affine/linalg lowering, CUDA/NVPTX kernel code-gen, MLIR.smt + Z3 runtime verification, and out-of-tree LLVM MachineScheduler plugins for MIR scheduling experiments, as well as LLVM ORC JIT compilation. In C++23, LLVM/MLIR 20.

Racket IR: Linklets
Co-designed the IR that ships in the main Racket distribution (see publications). Documented its design, and developed its formal semantics in my PhD dissertation.

Abstract Machine Interp: An executable theoretical model
Executable theoretical computation model that investigates switching between heap allocated continuations and native stack frames to reduce memory pressure in CESK machines. In PLT Redex.

Rax: A full-stack nanopass compiler from Racket to x86_64 ISA
Implemented all the passes (closure conversion, register allocation, code-gen, etc.), along with garbage collection. Developed optimizations, such as inlining, loop-invariant code motion, and proper tail calls. In Racket and C++14.

FARS: Functional Automated Reasoning System
A resolution/refutation theorem prover, for expressions in first-order predicate logic with equality. Used binary paramodulation, and forward and backward subsumption for equational deduction. In Racket.

HazirCevap (Witty): A closed-domain question-answering system for high school students
Government-funded large-scale question-answering system. M.Sc. thesis on NLP. Led the R&D team (3 faculty members, 4 grad students). Developed a Hidden Markov Random Field model for question analysis, and relevance metrics for information retrieval and response generation (see publications). In Python and JavaScript.

Selected Publications

- Flatt M., Derici C. Dybvig R. K., Keep A. et. al. "Rebuilding racket on chez scheme", ICFP'19
- Derici C. et. al. "A closed-domain question answering framework using reliable resources to assist students" Natural Language Engineering Journal'18
- Derici C. et. al. "Question analysis for a closed domain question answering system", CICLING'15
- Derici C. et. al. "Rule-based focus extraction in question answering systems", IEEE SIU'14
- Başar R. E., Derici C., and Şenol Ç. "World With Web: A compiler from world applications to JavaScript". Scheme and Functional Programming Workshop'09

Awards & Scholarships

- Scholarship and award for a project on teaching natural languages to hearing impaired, 2014.
- Full Scholarship for PhD, 2015-2025
- Full Scholarship for MSc, 2012
- Full Scholarship for BSc, 2005-2010
